In the next section, we briefly review the BM25 algorithm, which is a next-generation enhanced TF/IDF algorithm.

THE OKAPI BM25 ALGORITHM

Three main factors are involved in associating a relevancy score with the results: term frequency (TF), inverse document frequency (IDF), and field-length norm. Let's look at these factors briefly and learn how they affect relevancy.

Term frequency represents the number of times the search word appears in the current document's field. If we search for a word in a title field, the number of times the word appears is denoted by the term frequency variable. The higher the frequency, the higher the score.

Say, for example, that we are searching for the word *Java* in a title field across three documents. When indexing, we created the inverted index with similar information: the word, the number of times that word appears in that field (in a document), and the document IDs. We can create a table with this data, as table 3.6 shows.

Table 3.6 Term frequency for a search keyword

Title	Frequency	Doc ID
Mastering Java: Learning Core Java and Enterprise Java With Examples	3	25
Effective Java	1	13
Head First Java	1	39

Java appears three times in the document with ID 25 and one time in the other two documents. Because the search word appears more often in the first document (ID 25), it is logical to consider that document our favorite. Remember, the higher the frequency, the greater the relevance.

While this number seems to be a pretty good indication of the most relevant document in our search result, it is often not enough. Another factor, *inverse document frequency*, when combined with TF, produces improved scores.

The number of times the search word appears across the whole set of documents (i.e., across the whole index) is the *document frequency*. If the document frequency of a word is high, we can deduce that the search word is common across the index. If the word appears multiple times across all the documents in an index, it is a common term and, accordingly, not that relevant.

Words that appear often are not significant: words like *a*, *an*, *the*, *it*, and so forth are common in natural language and hence can be ignored. The inverse of the document frequency (*inverse document frequency*) provides a higher significance for uncommon words across the whole index. Hence, *the higher the document frequency, the lower the relevancy, and vice versa.* Table 3.7 shows the relationship between word frequency and relevance.